

SOLAR OVEN FUN!

A Grade 3 Science Lesson on Renewable Energy

Have you ever stepped inside a car on a hot day? Or walked barefoot across a sunny terrace? It feels super hot! That happens because the sun's amazing energy gets trapped inside the car—just like magic. Today, we are going to learn how to capture that exact same heat using a solar oven we build ourselves!

1. What is Renewable Energy?

Renewable energy means power that comes from nature and **never runs out**—just like the sun! Think about your phone battery: it runs down and needs to be plugged in to charge. But the sun? It shines every single day without ever needing a recharge! The sun is a completely free and forever source of energy. We can use its bright light and warm heat to do wonderful things, like cooking delicious treats!

2. Meet the 3 Magical Parts of a Solar Oven

Before we start building, let's meet the three special team members that make our solar oven work:

1. Aluminium Foil Flap

"The Mirror Friend"

It bounces and directs bright sunbeams straight into the oven box to make it super warm!

2. Plastic Wrap Cover

"The Magic Trapdoor"

It lets the glorious sunlight pass right through but keeps all the cozy heat securely trapped inside!

3. Black Paper Lining

"The Heat Sponge"

It soaks up all the sun's energy to warm up our food perfectly—just like real magic!

3. Step-by-Step: Let's Build Our Solar Oven!

Follow these three easy steps to put your solar oven together:

Step 1: Line the Box (The Heat Sponge)

Stick your black paper inside the bottom and walls of your cardboard box. This forms our Heat Sponge that stands ready to soak up all the warm sunshine!

Step 2: Seal the Top (The Magic Trapdoor)

Stretch clear plastic wrap tightly over the opening of the box and seal it well. Our Magic Trapdoor will allow light inside while trapping all the warm air.

Step 3: Add the Flap (The Mirror Friend)

Create a flap on the lid covered in aluminium foil. Position and angle the foil flap carefully so it catches the sunlight and bounces it directly down into the box!

4. The Awesome Science Behind the Oven

Bouncing Light (Reflection)

When sunlight hits a highly shiny surface like our aluminium flap, it bounces off—just like a ball bouncing off a wall! This is called **reflection**. It works exactly like a shiny steel thali bouncing sunbeams around the room.

Dark Colors Soak Up Heat (Absorption)

Dark colors, especially black, absorb heat much better than light colors. That is why wearing a black sweater during a hot sunny April afternoon makes you super sweaty! The black paper lining acts as our ultimate heat sponge.

Try This Quick Science Experiment: Touch a dark black surface and a clean white surface that have both been sitting in the sun. Which one feels significantly hotter to your hand?

5. Mini-Chef Recipe Time!

Sunshine Melts

What you can make today:

- Melted cheese on yummy sliced bread
- Melted chocolate chips over crisp Marie biscuits

Instructions: Place your treats inside the oven. Remember: use **only sunlight**—absolutely no gas stoves and no matches are allowed! Place your solar oven out in bright, open sunlight, wait patiently, and watch the magic cooking happen.

6. Eco-Hero Choices & Safety Rules

DO:

Use your solar oven on bright, beautifully sunny days—like a perfectly clear Sunday noon when the sun shines at its absolute strongest! More sunshine means more delicious heat inside.

DON'T:

Never put plastic toys or plastic bags inside your oven because they will melt! Avoid cloudy or rainy monsoon days too, as the hidden sun won't provide enough energy to cook.



IMPORTANT SAFETY RULES FOR ECO-CHEFS:

1. NEVER look directly at the sun!
2. ALWAYS run your experiments with an adult helper nearby.